Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 2

Date: 24-04-2026

React JS based web application for Ticket Booking of an Internal Department Event

SDG: SDG No – SDG 8 – Decent Work and Economic Growth

Presented By:

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INTRODUCTION

- **Background of the problem:** Historically, managing university department events, such as technical fests and academic seminars, has relied heavily on fragmented communication channels like localized WhatsApp groups, unorganized email chains, and physical paper forms. This outdated methodology is highly inefficient, creating severe communication bottlenecks and a disjointed experience for the student body.

- **Why this project is important:** EventSphere introduces a complete digital transformation to this traditional workflow. It provides a comprehensive, centralized full-stack web application designed to govern the entire event lifecycle securely. By utilizing a modern React JS frontend paired with a robust Spring Boot backend, the system delivers a seamless, highly responsive user experience tailored specifically for academic environments.

- **Real-world relevance:** In today's digital-first educational landscape, universities require highly organized, automated systems to handle massive student volumes safely and efficiently. This system directly addresses the logistical friction of campus management, replacing chaos with streamlined data.

- **Brief overview of the system:** The platform features dedicated, role-based dashboards that cater specifically to the distinct administrative privileges of Administrators, Faculty Coordinators, and Students. The entire application is elegantly wrapped in a visually appealing dark glass-morphism interface, providing a premium, modern aesthetic.

PROBLEM STATEMENT

Fragmented Discovery: Students currently lack a unified, reliable digital platform to discover upcoming department events. Because of this, they often miss out on critical academic seminars or workshops due to a heavy reliance on scattered physical bulletin boards or informal, unreliable messaging networks.

Manual Bottlenecks: On the administrative side, managing manual ticket sales and physical cash collection creates massive logistical friction. Furthermore, physical ticket validation leads to long, unmanageable queues at venue doors and introduces significant human error into the check-in process.

Data Integrity Issues: Traditional registration methods frequently result in critical logistical failures. Administrators constantly battle issues such as accidental overbooking, lost registration data, and duplicate student entries for the exact same event, which entirely skews capacity planning.

Lack of Analytics: Event coordinators and faculty members are currently forced to operate blindly. They lack real-time attendance analytics to track participation and do not have automated tools to instantly notify registered attendees of sudden schedule changes or venue updates.

Security Gaps: There is a pressing need for an automated system with strict, role-based access control to securely separate student viewing rights from powerful faculty and administrative management privileges.

OBJECTIVE(S)

- **Design and Develop:** The primary objective is to architect and build a full-stack, single-page web application using React JS (Vite) and Java Spring Boot. This robust platform will securely manage and entirely automate the lifecycle of university department events from inception to completion.

- **Implement Contactless Systems:** A critical technical goal is to implement a highly secure, contactless QR-code check-in system. By utilizing browser-based scanning technology (html5-qrcode), the application aims to completely eliminate physical queues at event venues, allowing for rapid, digital ticket validation.

- **Automate Communications:** The system will drastically reduce administrative overhead by dispatching automated email confirmations containing secure QR attachments directly to registered users immediately upon a successful database booking.

- **Improve Booking Integrity:** The project aims to engineer robust, transactional backend logic that strictly prevents race conditions. This specifically includes a definitive duplicate-registration guard to ensure fairness and strictly accurate capacity limits.

- **Evaluate System Performance:** The final objective is to rigorously test the application's overall ability to handle concurrent ticket purchases seamlessly while maintaining strictly accurate, real-time ticket inventory updates across the relational database tables.

PROPOSED SYSTEM

- **Digital Ecosystem:** The proposed solution, EventSphere, is a secure, highly automated event management ecosystem. It functions as a modern single-page application (SPA) that completely digitizes and optimizes the traditional paper-based university workflow.

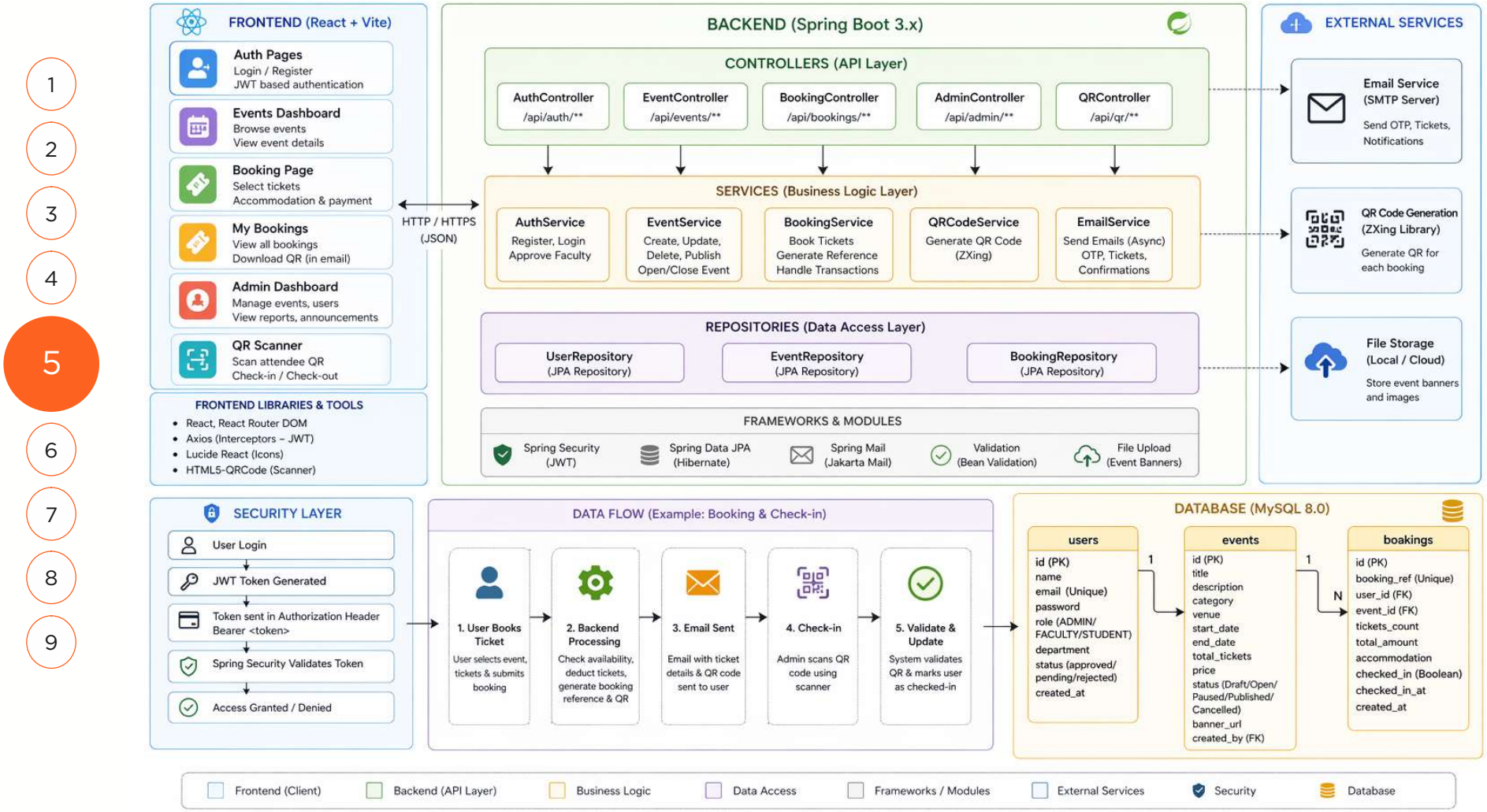
- **Contactless Ticketing:** It actively solves physical ticketing issues by dynamically generating unique, Base64-encoded QR codes for every successful student booking. These secure digital tickets are automatically dispatched to the student's university email, preventing loss and forgery entirely.

- **Modern Authentication:** The system provides a significant advantage over legacy university portals by transitioning to stateless JWT (JSON Web Token) authentication. This ensures high performance, highly secure session management, and immediate readiness for cloud-scale deployment.

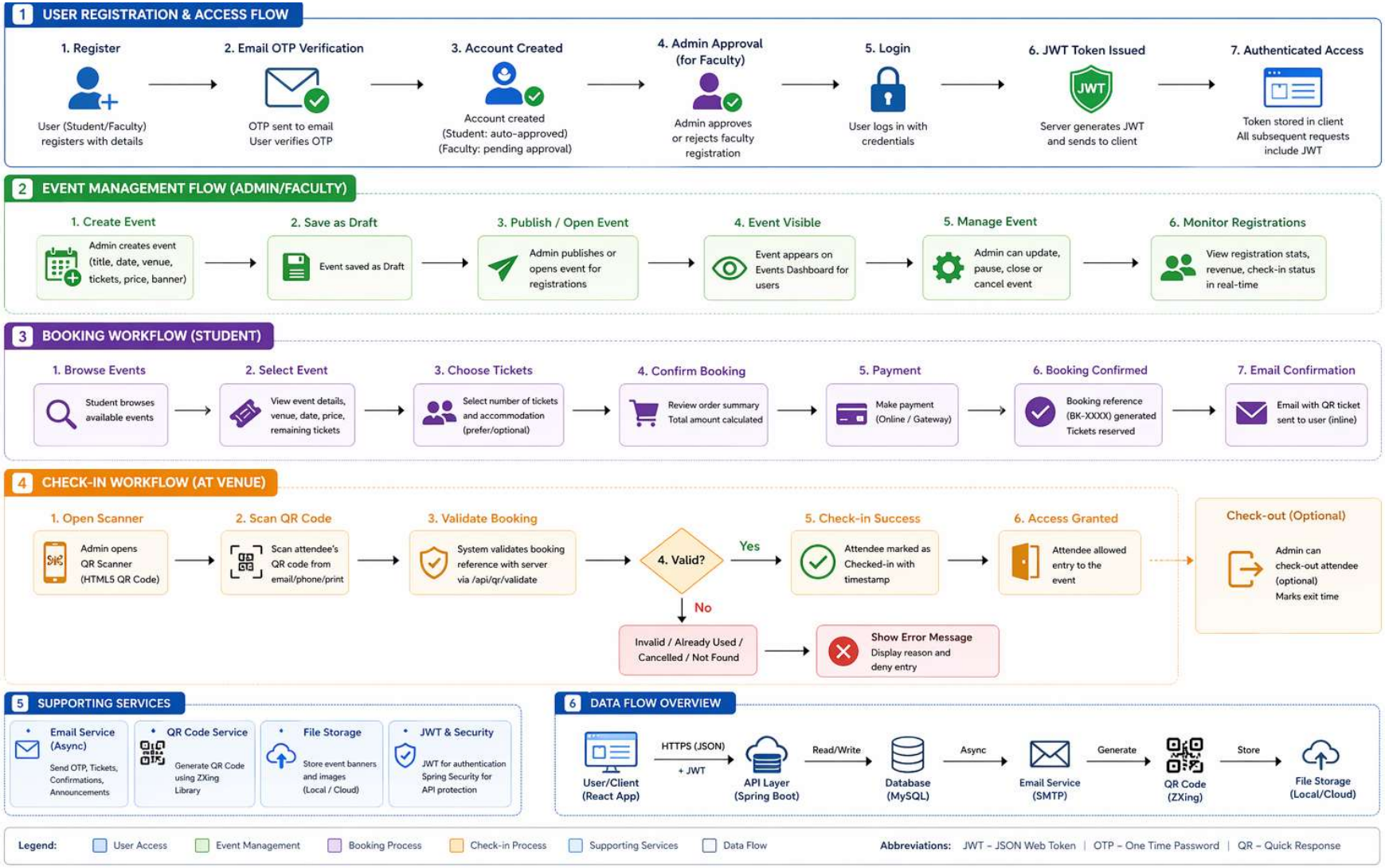
- **End-to-End Pipeline:** EventSphere delivers a comprehensive digital pipeline: high-level administrators can effortlessly publish new events, students can dynamically filter and book tickets, and faculty can instantly validate attendees.

- **Instant Validation Integration:** It features a built-in browser webcam scanner designed specifically for administrators. This specialized tool enables instant, highly accurate check-ins directly at the venue doors, drastically improving crowd flow and overall event security.

PROPOSED SYSTEM ARCHITECTURE



WORKFLOW


1 USER REGISTRATION & ACCESS FLOW
1. Register
User (Student/Faculty) registers with details
2. Email OTP Verification
OTP sent to email User verifies OTP
3. Account Created
Account created (Student: auto-approved) (Faculty: pending approval)
4. Admin Approval (for Faculty)
Admin approves or rejects faculty registration
5. Login
User logs in with credentials
6. JWT Token Issued
JWT
Server generates JWT and sends to client
7. Authenticated Access
Token stored in client All subsequent requests include JWT
2 EVENT MANAGEMENT FLOW (ADMIN/FACULTY)
1. Create Event
Admin creates event (title, date, venue, tickets, price, banner)
2. Save as Draft
Event saved as Draft
3. Publish / Open Event
Admin publishes or opens event for registrations
4. Event Visible
Event appears on Events Dashboard for users
5. Manage Event
Admin can update, pause, close or cancel event
6. Monitor Registrations
View registration stats, revenue, check-in status in real-time
3 BOOKING WORKFLOW (STUDENT)
1. Browse Events
Student browses available events
2. Select Event
View event details, venue, date, price, remaining tickets
3. Choose Tickets
Select number of tickets and accommodation (prefer/optional)
4. Confirm Booking
Review order summary Total amount calculated
5. Payment
Make payment (Online / Gateway)
6. Booking Confirmed
Booking reference (BK-XXXX) generated Tickets reserved
7. Email Confirmation
Email with QR ticket sent to user (inline)
4 CHECK-IN WORKFLOW (AT VENUE)
1. Open Scanner
Admin opens QR Scanner (HTML5 QR Code)
2. Scan QR Code
Scan attendee's QR code from email/phone/print
3. Validate Booking
System validates booking reference with server via /api/qr/validate
4. Valid?
Yes
No
5. Check-in Success
Attendee marked as Checked-in with timestamp
6. Access Granted
Attendee allowed entry to the event
Check-out (Optional)
Admin can check-out attendee (optional) Marks exit time
Invalid / Already Used / Cancelled / Not Found
Show Error Message
Display reason and deny entry
5 SUPPORTING SERVICES
Email Service (Async)
Send OTP, Tickets, Confirmations, Announcements
QR Code Service
Generate QR Code using ZXing Library
File Storage
Store event banners and images (Local / Cloud)
JWT & Security
JWT for authentication Spring Security for API protection
6 DATA FLOW OVERVIEW
User/Client (React App)
HTTPS (JSON) + JWT
API Layer (Spring Boot)
Read/Write
Database (MySQL)
Async
Email Service (SMTP)
Generate
QR Code (ZXing)
Store
File Storage (Local/Cloud)
Legend: User Access | Event Management | Booking Process | Check-in Process | Supporting Services | Data Flow
Abbreviations: JWT – JSON Web Token | OTP – One Time Password | QR – Quick Response

TECHNOLOGIES USED

- **Frontend Technologies:** The presentation layer is built utilizing modern React JS and the Vite build tool, supported by standard HTML5 and CSS3. The interface features a highly responsive, custom dark glass-morphism UI designed specifically to provide a premium, visually engaging user experience.

- **Backend API Framework:** The server environment relies heavily on Java 17 and the enterprise-grade Spring Boot framework. It utilizes Spring MVC for endpoint routing and the mature Spring Security framework to gracefully handle JWT-based stateless authentication.

- **Database Storage:** A relational MySQL 8.0+ database is successfully implemented. It is combined seamlessly with Spring Data JPA (Hibernate) for highly efficient object-relational mapping and strict transactional integrity across all tables.

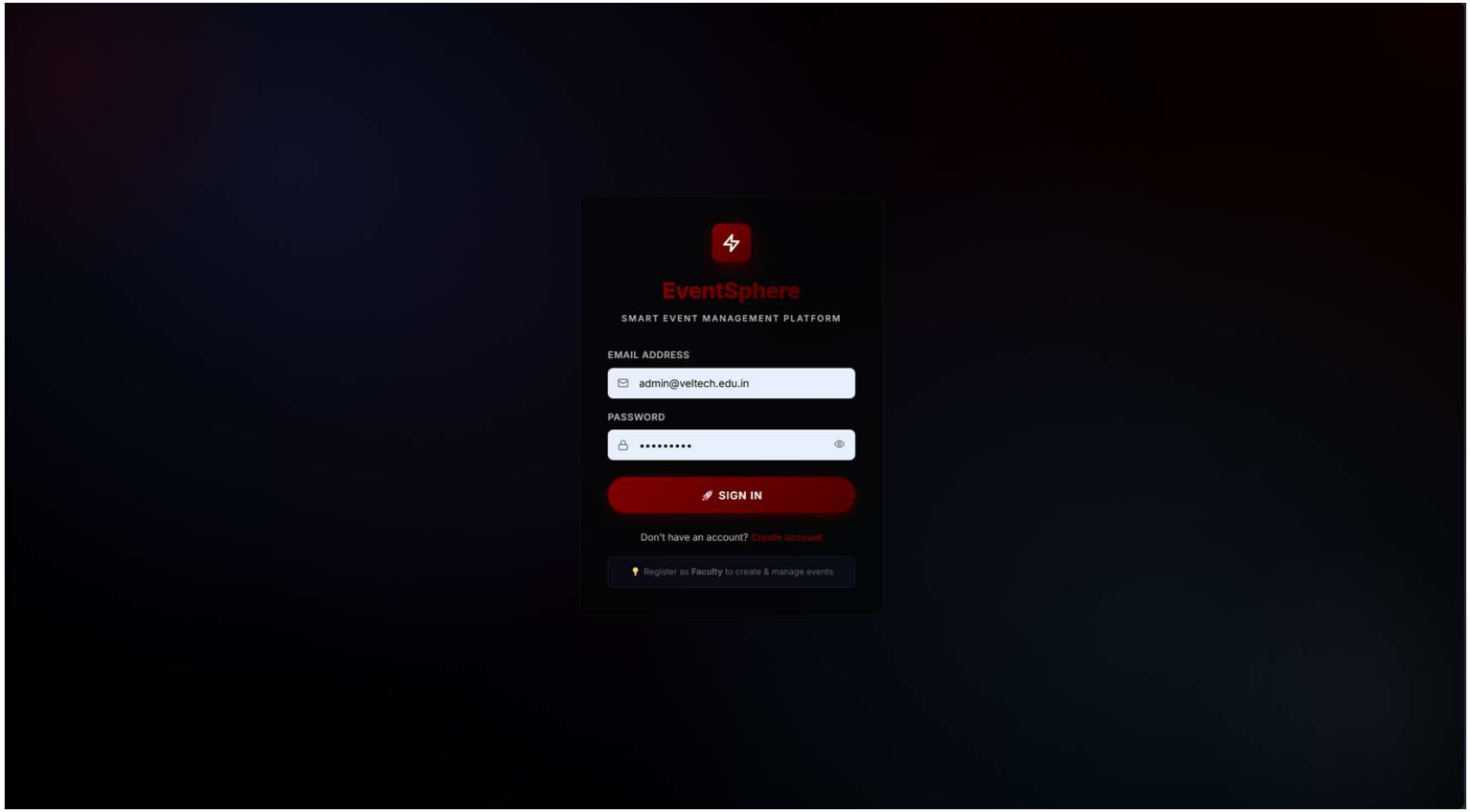
- **Key Libraries & Integrations:** Critical project dependencies include Axios for handling asynchronous client-server HTTP requests safely, the robust ZXing library for highly reliable server-side QR generation, and the html5-qrcode library for native browser webcam scanning.

- **System Requirements:** To properly run and develop the application locally, a standard PC/Laptop is required, alongside JDK 17+, Node.js version 18+, NPM 9+, and a locally running MySQL Server instance on port 3306.

MODULE DESCRIPTION

- **Module 1: User Identity & Security:** This foundational backend module is entirely responsible for handling secure, stateless JWT generation during the login phase. It utilizes strong BCrypt password hashing to protect credentials and strictly enforces protected route access across all distinct user roles (Admin, Faculty, Student).
- **Module 2: Event & Booking Engine:** This core module contains the application's primary business logic. It allows administrators to execute full CRUD (Create, Read, Update, Delete) operations on events. Crucially, it actively enforces a highly strict "duplicate-registration guard" during student ticket checkout to maintain flawless inventory accuracy.
- **Module 3: QR Validation & Check-In:** Bridging the digital interface directly with physical event attendance, this module dynamically generates secure, non-guessable booking references. It provides the dedicated web scanner interface, safely allowing administrators to execute rapid, real-time venue check-ins using standard device webcams.
- **Module 4: Admin Analytics & Communication:** Acting as the central command hub for the university, this module continuously aggregates total ticket sales data and calculates precise capacity percentages. It also manages faculty coordinator assignments and powers the automated campaign email broadcaster for instant, mass student communication.

IMPLEMENTATION SCREENSHOTS



IMPLEMENTATION SCREENSHOTS

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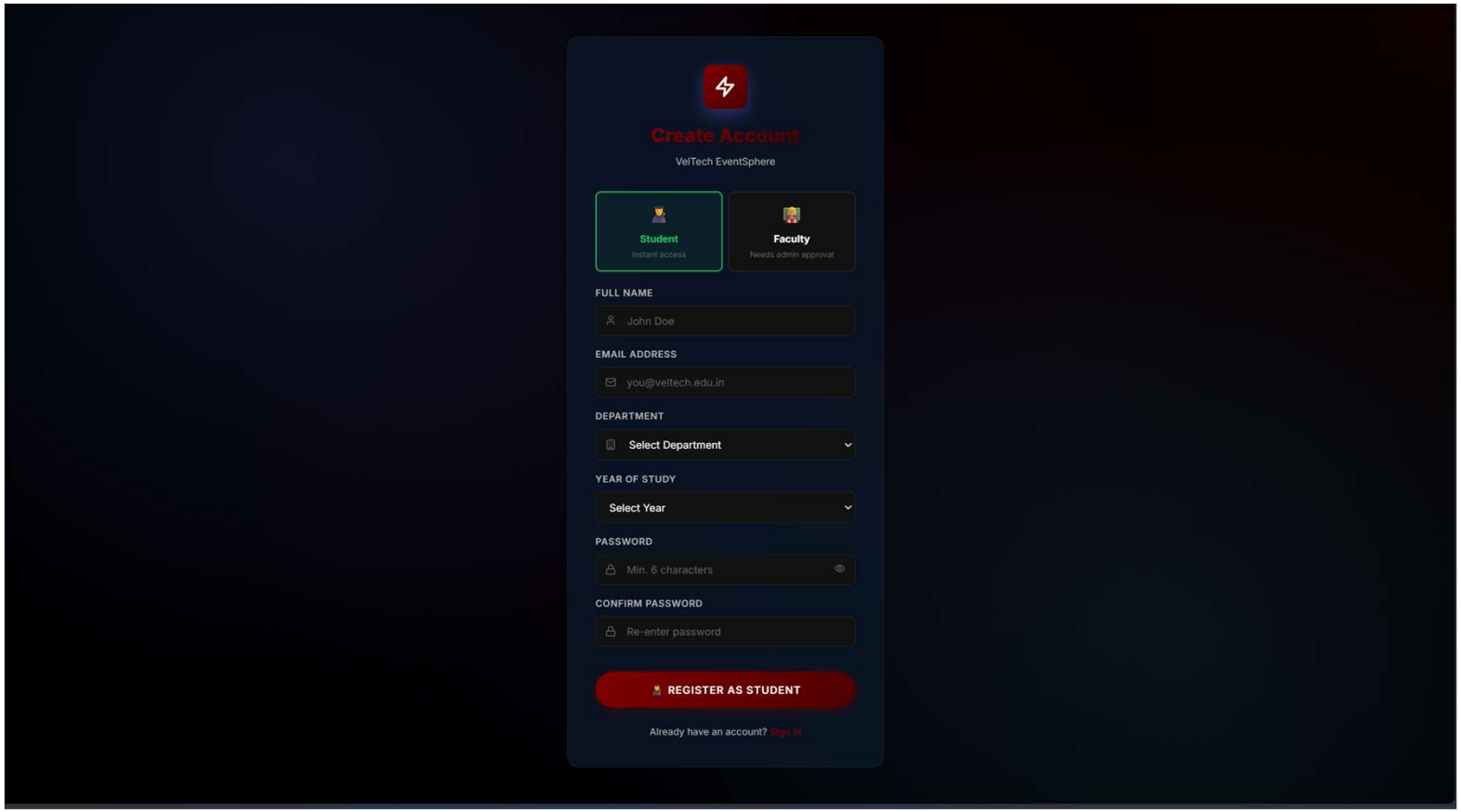

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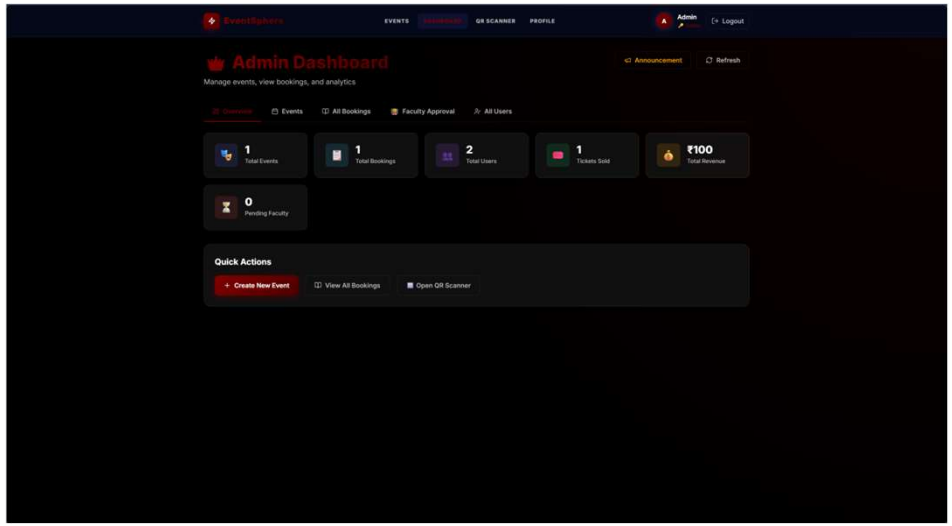

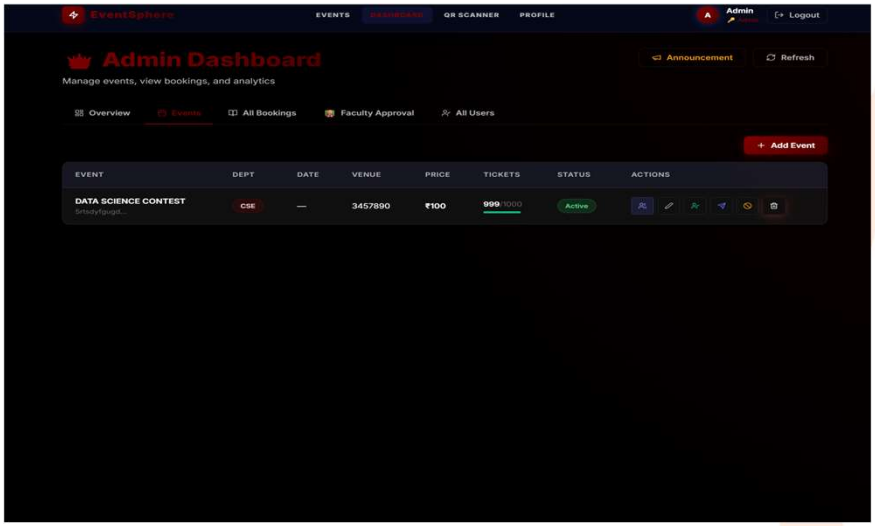

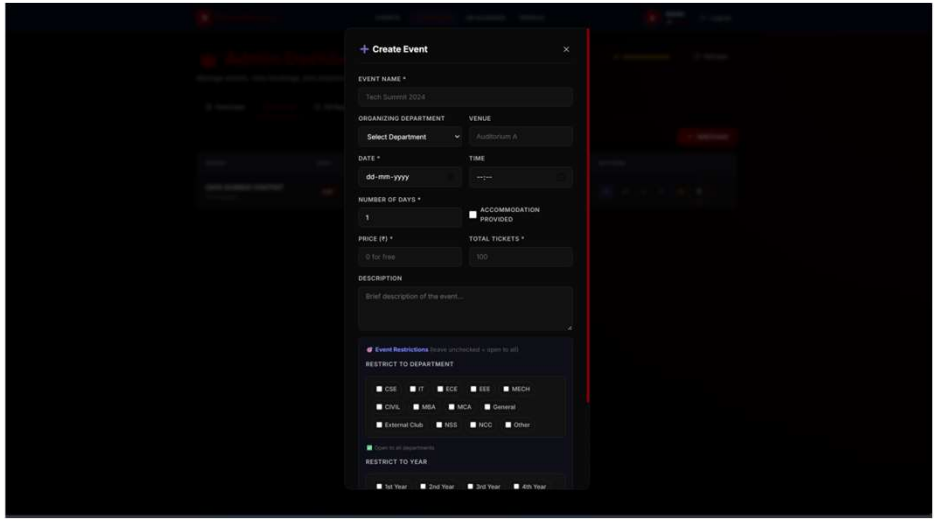

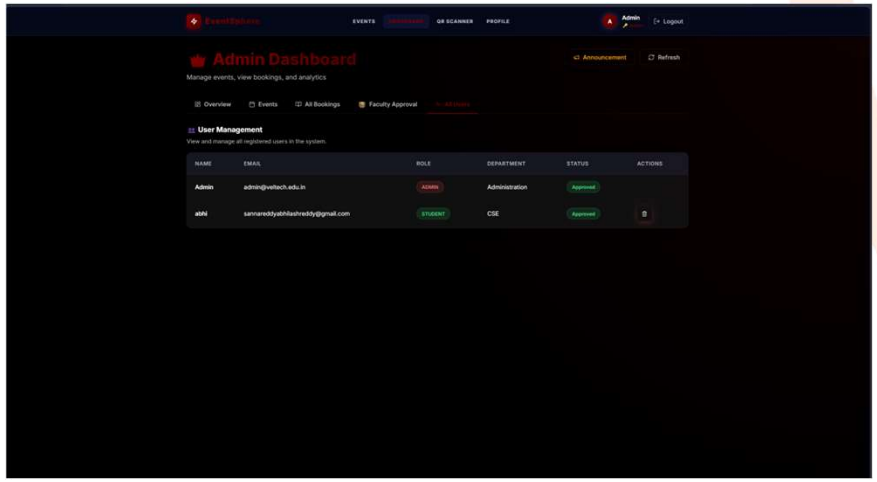

IMPLEMENTATION SCREENSHOTS

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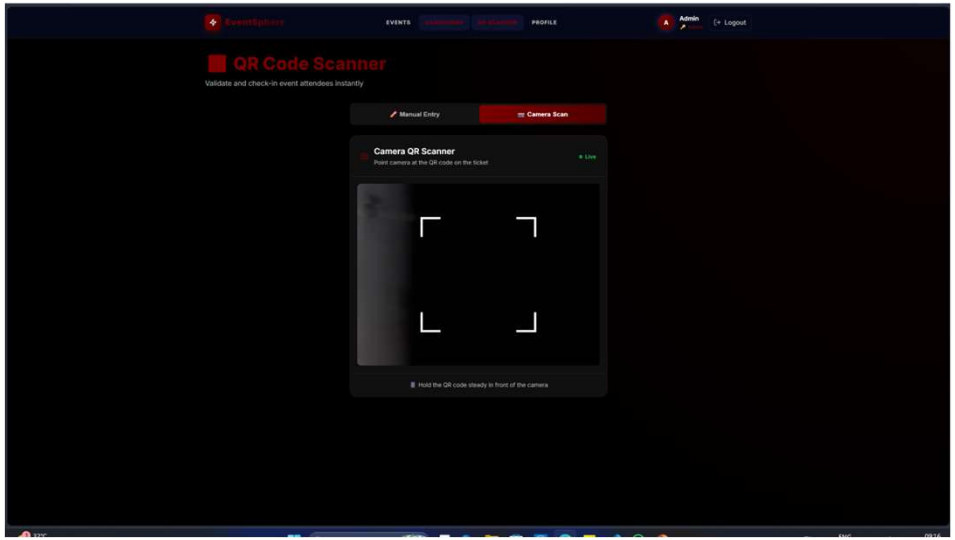

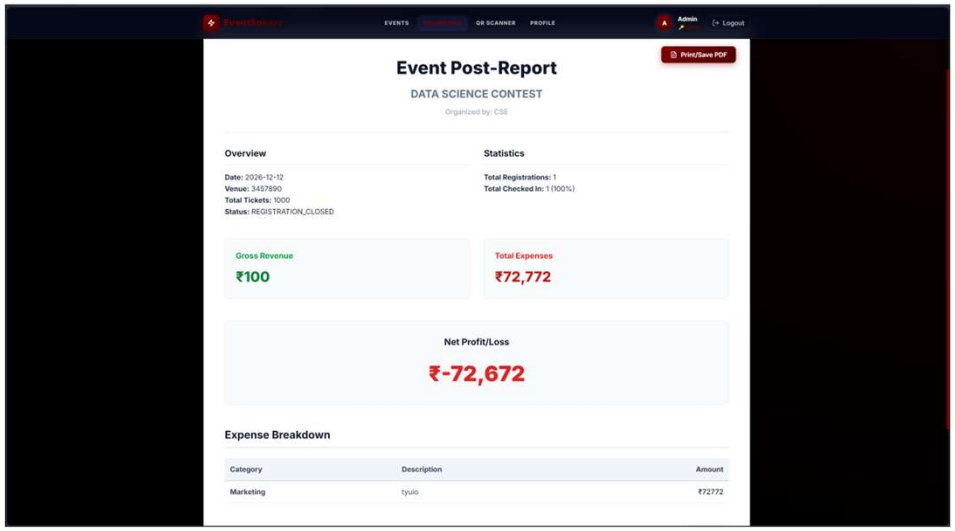

ADVANTAGES & LIMITATIONS

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Advantages (Integrity):
Guarantees real-time ticket accuracy through robust transactional database locks and actively prevents duplicate registrations.

Advantages (Security):
Secures user data with stateless JWT authentication and saves faculty hours of manual work through automated email workflows.

Limitations (Email Constraints):
Currently relies on a basic Gmail App Password, which lacks the high scalability required for massive campus-wide broadcasts.

Limitations (Hosting Setup):
The application is configured for local deployment; moving it to a cloud provider like AWS would be necessary for high availability.

Limitations (Future Scope):
Features like bulk expense data export and custom UI theme toggles are planned but not yet fully implemented on the backend.

Thank you